

■ Magic Number Lab

Large Numbers — Place Value to Lakhs

A Maths Lesson for Young Wizards

■ Place Value

■ 6-Digit Numbers

■ Expanded Form

■ Comparing Numbers

■ What You Will Learn

- ◆ Name the 6 places in the Indian place-value system
- ◆ Find the place value of any digit in a large number
- ◆ Write numbers in expanded form and standard form
- ◆ Compare and order 5- and 6-digit numbers

■ Did You Know?

Every area in India has a 6-digit PIN code! For example, a PIN code like **600001** is actually a 6-digit large number. That means PIN codes go all the way up to the Lakhs place! ■■

■ CONCEPT 1 — The Indian Place Value System

In India, we read big numbers using a special system. Numbers are grouped with commas in a unique Indian way.

The 6 Places (right to left):

Place	Short Form	Value	Example
Ones	O	1	6
Tens	T	10	10
Hundreds	H	100	100
Thousands	T	1,000	1,000
Ten-Thousands	TT	10,000	10,000
Lakhs	L	1,00,000	1,00,000

■ CONCEPT 2 — The Place Value Chart

Here is the number **3,25,416** shown in the place value chart:

Lakhs (L)	Ten-Thousands (TT)	Thousands (T)	Hundreds (H)	Tens (T)	Ones (O)
3	2	5	4	1	6
3,00,000	20,000	5,000	400	10	6

■■ Indian Comma Rule

In India, we place commas differently from other countries:

First comma after the **3rd digit** from the right (after hundreds)

Second comma after the **5th digit** from the right (after ten-thousands)

Example: 325416 → **3,25,416** ✓

NOT like this: 325,416 ✗

■ CONCEPT 3 — Face Value vs Place Value

Digit 3 in 3,25,416	3 (always)	$3 \times 1,00,000 = \mathbf{3,00,000}$
Digit 2 in 3,25,416	2 (always)	$2 \times 10,000 = \mathbf{20,000}$
Digit 5 in 3,25,416	5 (always)	$5 \times 1,000 = \mathbf{5,000}$
Digit 4 in 3,25,416	4 (always)	$4 \times 100 = \mathbf{400}$
Digit 1 in 3,25,416	1 (always)	$1 \times 10 = \mathbf{10}$
Digit 6 in 3,25,416	6 (always)	$6 \times 1 = \mathbf{6}$

■ CONCEPT 4 — Expanded Form

Expanded form means writing a number as a **sum of place values**.

■ Worked Examples

Example 1: 4,56,302

$$= 4,00,000 + 50,000 + 6,000 + 300 + 0 + 2$$

= Four lakh fifty-six thousand three hundred two

Example 2: 7,08,050

$$= 7,00,000 + 0 + 8,000 + 0 + 50 + 0$$

= Seven lakh eight thousand fifty

Example 3: 1,00,001

$$= 1,00,000 + 0 + 0 + 0 + 0 + 1$$

= One lakh and one

■■ CONCEPT 5 — Comparing Large Numbers

■ Rules for Comparing

Rule 1: The number with **MORE digits** is always greater.

Example: $1,00,000 > 99,999$ (6 digits $>$ 5 digits)

Rule 2: If both numbers have the **SAME** number of digits, compare digit by digit from **left to right**.

Example: Compare 4,52,310 and 4,61,200

Both have 6 digits. First digits: $4 = 4$ (same)

Second digits: $5 < 6$, so **4,52,310 $<$ 4,61,200**

Ascending Order = Smallest to Largest ■■

Descending Order = Largest to Smallest ■■

Example: 3,10,000 / 85,900 / 4,25,100 / 99,999

Ascending: $85,900 < 99,999 < 3,10,000 < 4,25,100$

Descending: $4,25,100 > 3,10,000 > 99,999 > 85,900$

⇒■■ PRACTICE TIME

Try these questions! Use your place value chart to help you.

Q1. Write the place value of the underlined digit: 6,35,201

Answer: 3 is in Ten-Thousands place $\rightarrow$ Place Value = 30,000

Q2. Write 5,08,314 in expanded form.

Answer: $5,00,000 + 8,000 + 300 + 10 + 4$

Q3. Write the number name for 2,40,506.

Answer: Two lakh forty thousand five hundred six

Q4. Which is greater: 7,25,000 or 7,52,000?

Answer: $7,52,000 > 7,25,000$ (compare digit 2: $2 < 5$)

Q5. What is the successor of 9,99,999?

Answer: 10,00,000 (Ten Lakh — new place!)

Q6. Arrange in ascending order: 45,200 / 4,52,000 / 4,05,200 / 45,002

Answer: $45,002 < 45,200 < 4,05,200 < 4,52,000$

Q7. A mela had 3,50,000 visitors last year. Write this in expanded form.

Answer: $3,00,000 + 50,000 = 3,50,000$

Q8. The face value of 8 in 8,14,302 is ____ and place value is ____.

Answer: Face value = 8; Place value = 8,00,000

Q9. Write in standard form: 4,00,000 + 30,000 + 500 + 6

Answer: 4,30,506

Q1 A PIN code is 600041. Which digit is in the Lakhs place?
0.

Answer: Digit 6 is in the Lakhs place (value = 6,00,000)

■ Today I Learned...

Reflection Box — Fill in your own words!

1. The 6 places in the Indian system are:

_____ , _____ , _____ , _____ , _____ , _____

2. The place value of digit 5 in 5,32,100 is:

3. 3,06,200 in expanded form is:

4. One thing I found easy today: _____

5. One thing I want to practise more: _____

■■ Exit Ticket — 3 Quick Questions

Tick the correct answer:

1. How many digits does **1 Lakh** have?

■ 5 digits ■ 6 digits ■ 7 digits

2. The place value of 4 in 4,12,305 is:

■ 4,000 ■ 40,000 ■ 4,00,000

3. Which is the correct Indian notation for three lakh twenty-five?

■ 300,025 ■ 3,00,025 ■ 30,025

■ Keep Practising — You Are a Place Value Wizard! ■